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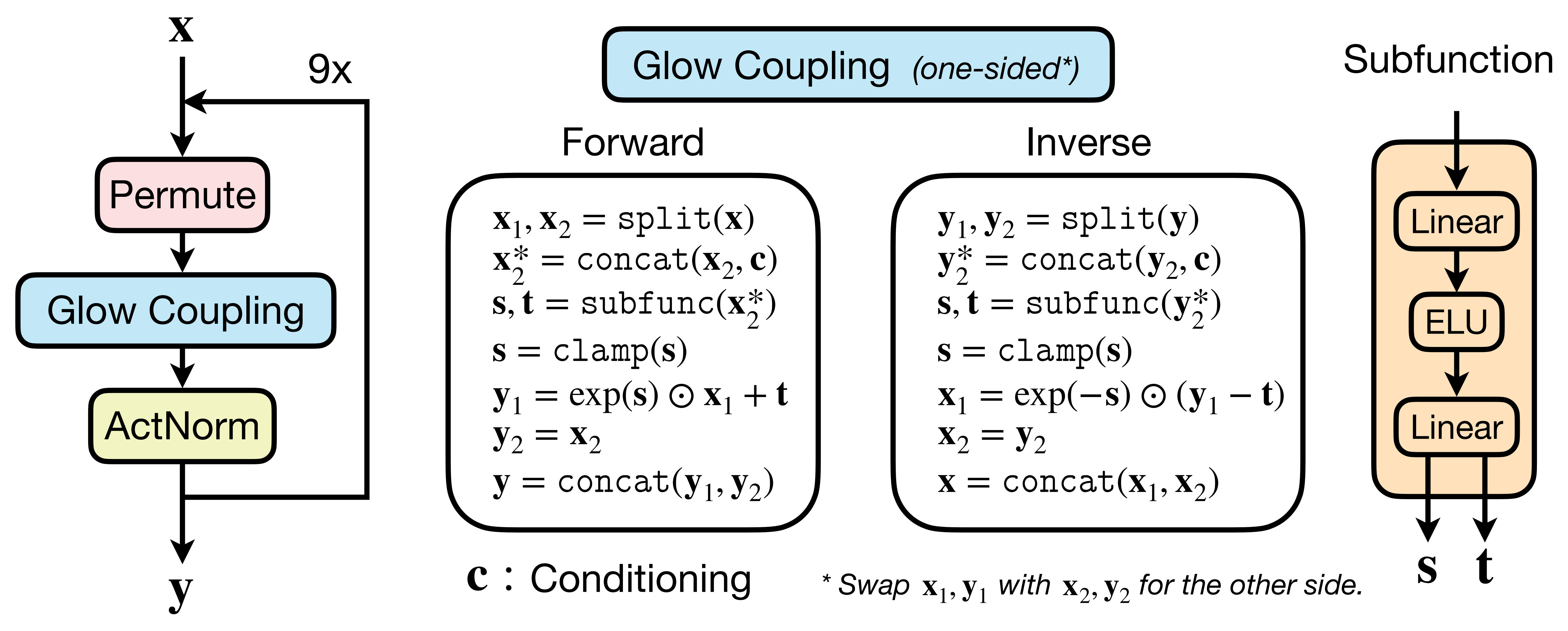
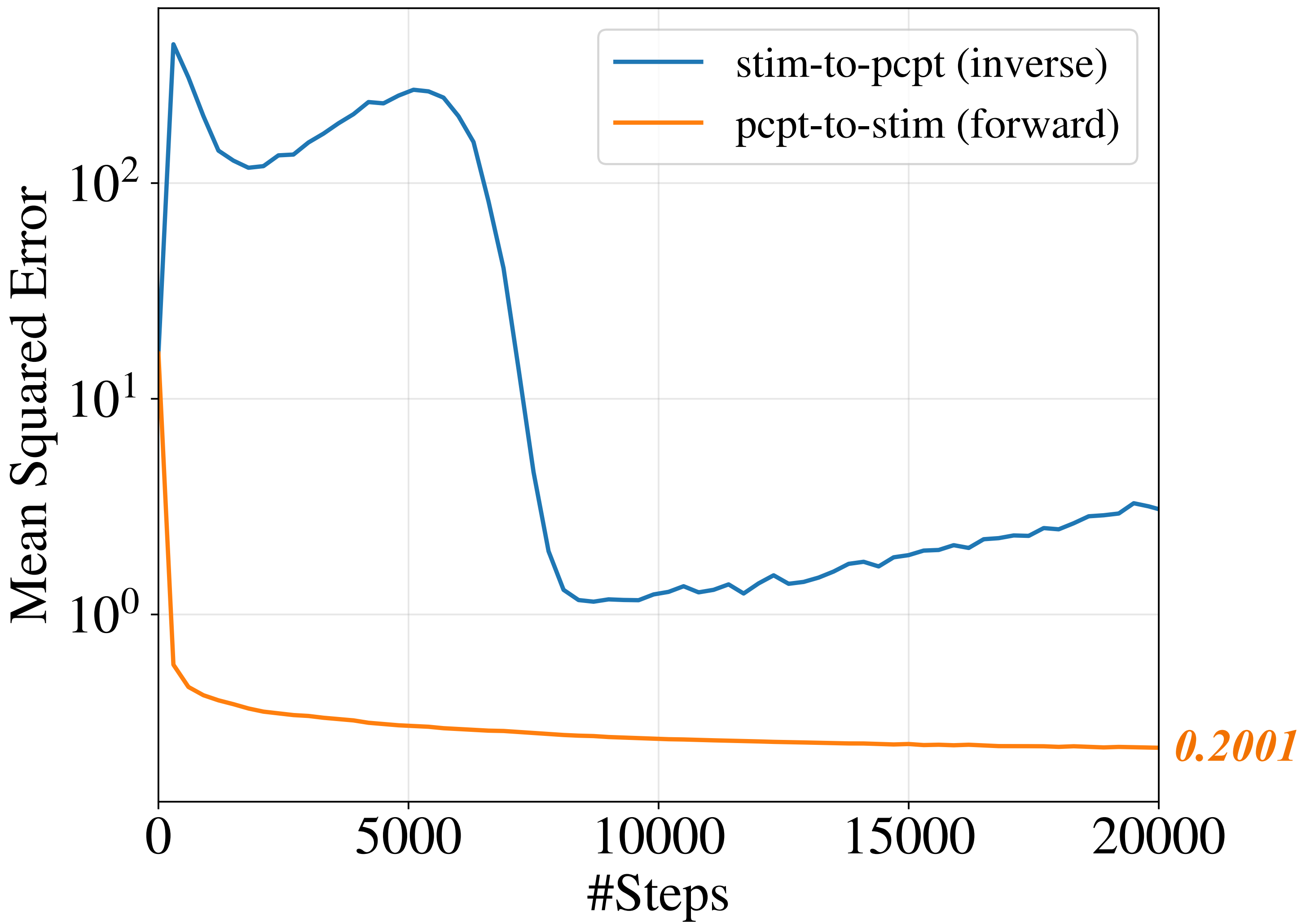
27

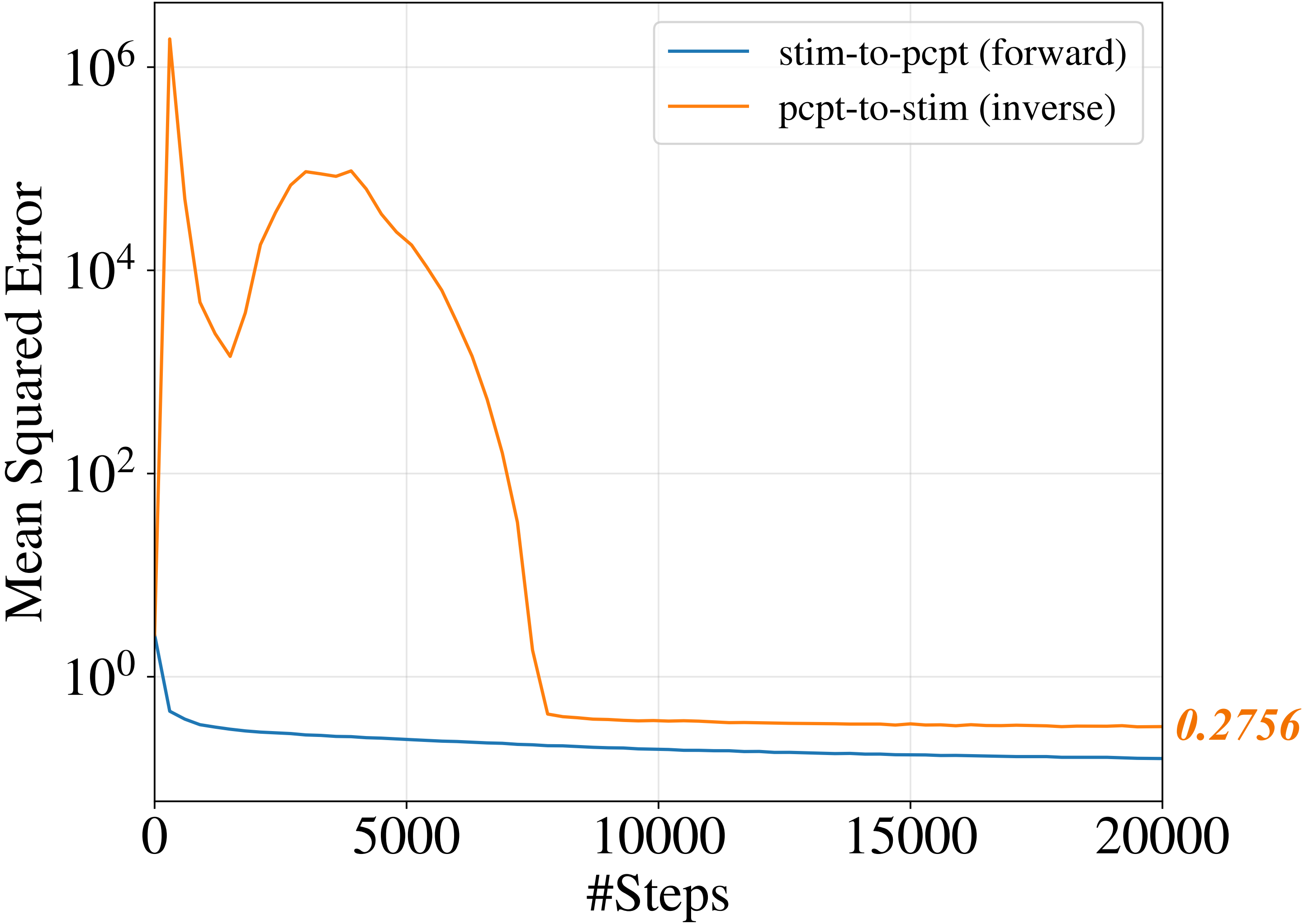
Invertible Neural Networks

[Wu et al., 2024b]



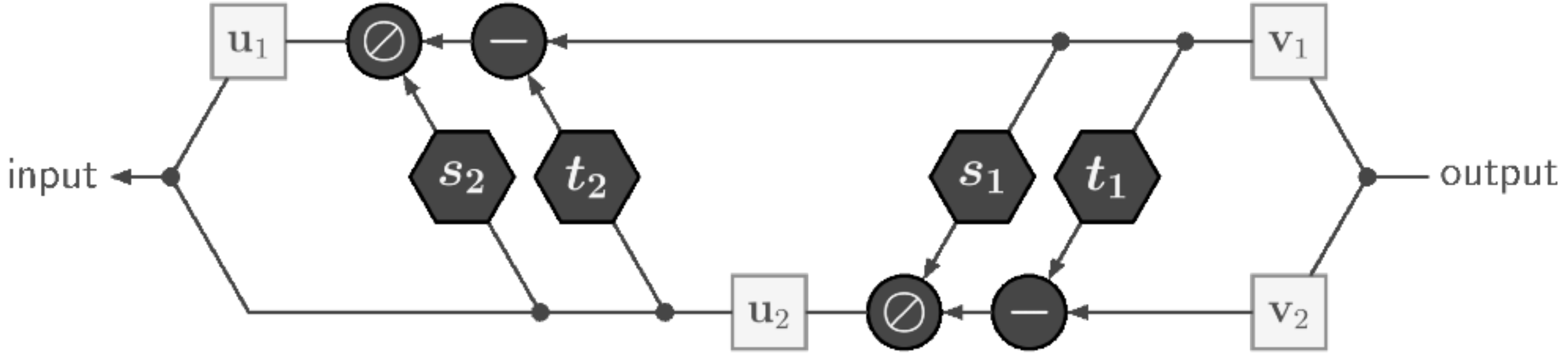
x
9x
Permute
Glow Coupling
ActNorm
y
Glow Coupling (one-sided*)
Forward
$\mathbf{x}_1, \mathbf{x}_2 = \texttt{split}(\mathbf{x})$
$\mathbf{x}_2^* = \texttt{concat}(\mathbf{x}_2, \mathbf{c})$
$\mathbf{s}, \mathbf{t} = \texttt{subfunc}(\mathbf{x}_2^*)$
$\mathbf{s} = \texttt{clamp}(\mathbf{s})$
$\mathbf{y}_1 = \exp(\mathbf{s}) \odot \mathbf{x}_1 + \mathbf{t}$
$\mathbf{y}_2 = \mathbf{x}_2$
$\mathbf{y} = \texttt{concat}(\mathbf{y}_1, \mathbf{y}_2)$
Inverse
$\mathbf{y}_1, \mathbf{y}_2 = \texttt{split}(\mathbf{y})$
$\mathbf{y}_2^* = \texttt{concat}(\mathbf{y}_2, \mathbf{c})$
$\mathbf{s}, \mathbf{t} = \texttt{subfunc}(\mathbf{y}_2^*)$
$\mathbf{s} = \texttt{clamp}(\mathbf{s})$
$\mathbf{x}_1 = \exp(-\mathbf{s}) \odot (\mathbf{y}_1 - \mathbf{t})$
$\mathbf{x}_2 = \mathbf{y}_2$
$\mathbf{x} = \texttt{concat}(\mathbf{x}_1, \mathbf{x}_2)$
$\mathbf{c}$: Conditioning
* Swap $\mathbf{x}_1, \mathbf{y}_1$ with $\mathbf{x}_2, \mathbf{y}_2$ for the other side.
Subfunction
Linear
ELU
Linear
s
t


stim-to-pcpt (inverse)
pcpt-to-stim (forward)
Mean Squared Error
10^2
10^1
10^0
0.2001
0
5000
10000
15000
20000
#Steps

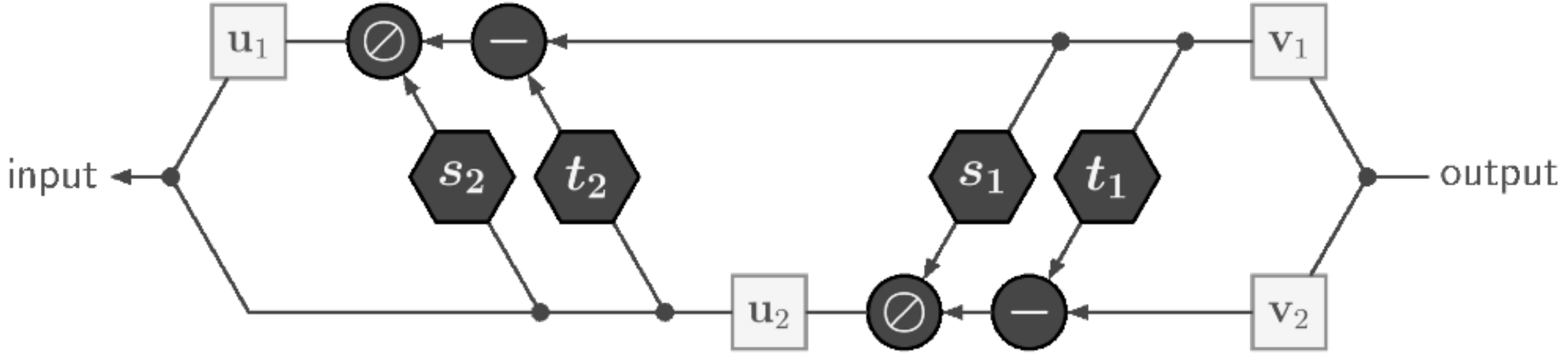


Mean Squared Error
10^6
10^4
10^2
10^0
0
5000
10000
15000
20000
#Steps
stim-to-pcpt (forward)
pcpt-to-stim (inverse)
0.2756

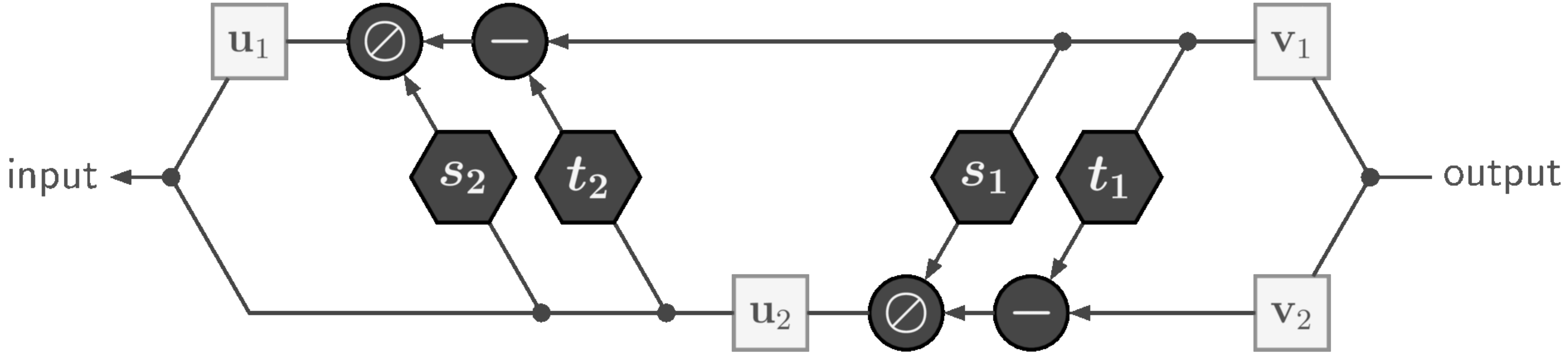
https://hci.iwr.uni-heidelberg.de/vislearn/inverse-problems-invertible-neural-networks/



$\mathbf{u}_1$
$\mathbf{v}_1$
input
s_2
t_2
s_1
t_1
output
$\mathbf{u}_2$
$\mathbf{v}_2$

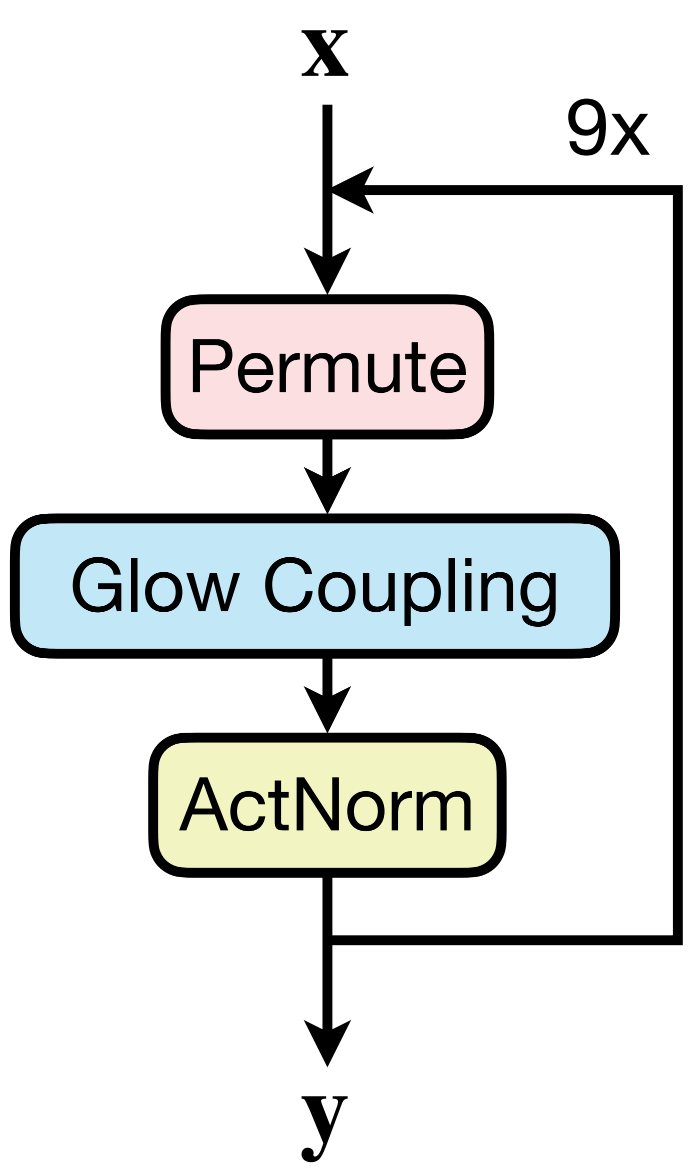


$\mathbf{u}_1$
$\mathbf{v}_1$
input
s_2
t_2
s_1
t_1
output
$\mathbf{u}_2$
$\mathbf{v}_2$



$\mathbf{u}_1$
$\mathbf{v}_1$
input
s_2
t_2
s_1
t_1
output
$\mathbf{u}_2$
$\mathbf{v}_2$

Invertible Neural Networks



Glow Coupling *(one-sided*)*

Forward

$$
\begin{aligned}
\mathbf{x}_1, \mathbf{x}_2 &= \mathtt{split}(\mathbf{x}) \\
\mathbf{x}_2^* &= \mathtt{concat}(\mathbf{x}_2, \mathbf{c}) \\
\mathbf{s}, \mathbf{t} &= \mathtt{subfunc}(\mathbf{x}_2^*) \\
\mathbf{s} &= \mathtt{clamp}(\mathbf{s}) \\
\mathbf{y}_1 &= \exp(\mathbf{s}) \odot \mathbf{x}_1 + \mathbf{t} \\
\mathbf{y}_2 &= \mathbf{x}_2 \\
\mathbf{y} &= \mathtt{concat}(\mathbf{y}_1, \mathbf{y}_2)
\end{aligned}
$$

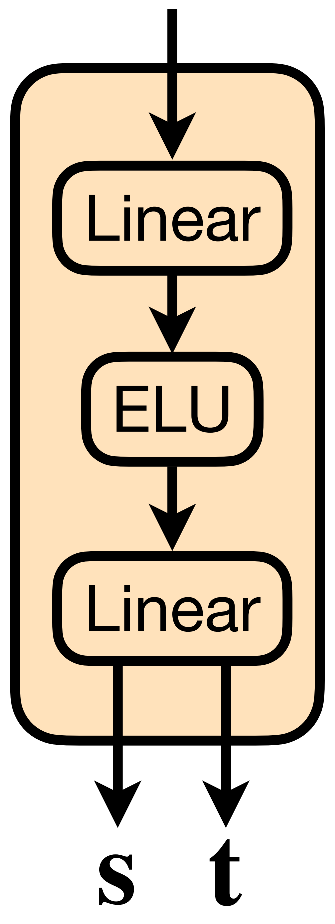
Inverse

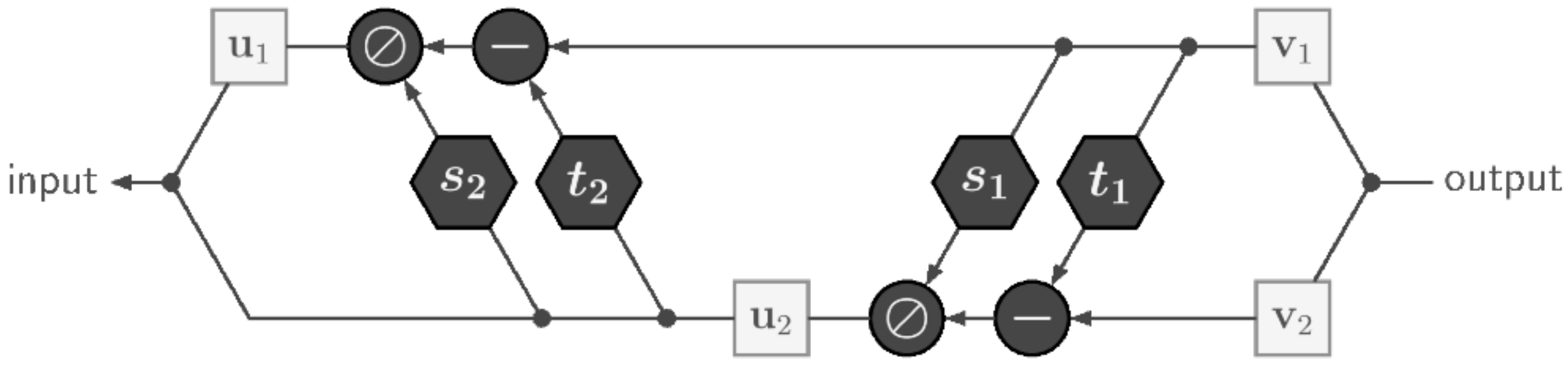
$$
\begin{aligned}
\mathbf{y}_1, \mathbf{y}_2 &= \mathtt{split}(\mathbf{y}) \\
\mathbf{y}_2^* &= \mathtt{concat}(\mathbf{y}_2, \mathbf{c}) \\
\mathbf{s}, \mathbf{t} &= \mathtt{subfunc}(\mathbf{y}_2^*) \\
\mathbf{s} &= \mathtt{clamp}(\mathbf{s}) \\
\mathbf{x}_1 &= \exp(-\mathbf{s}) \odot (\mathbf{y}_1 - \mathbf{t}) \\
\mathbf{x}_2 &= \mathbf{y}_2 \\
\mathbf{x} &= \mathtt{concat}(\mathbf{x}_1, \mathbf{x}_2)
\end{aligned}
$$

$\mathbf{c}$: Conditioning

* *Swap* $\mathbf{x}_1, \mathbf{y}_1$ *with* $\mathbf{x}_2, \mathbf{y}_2$ *for the other side.*

Subfunction





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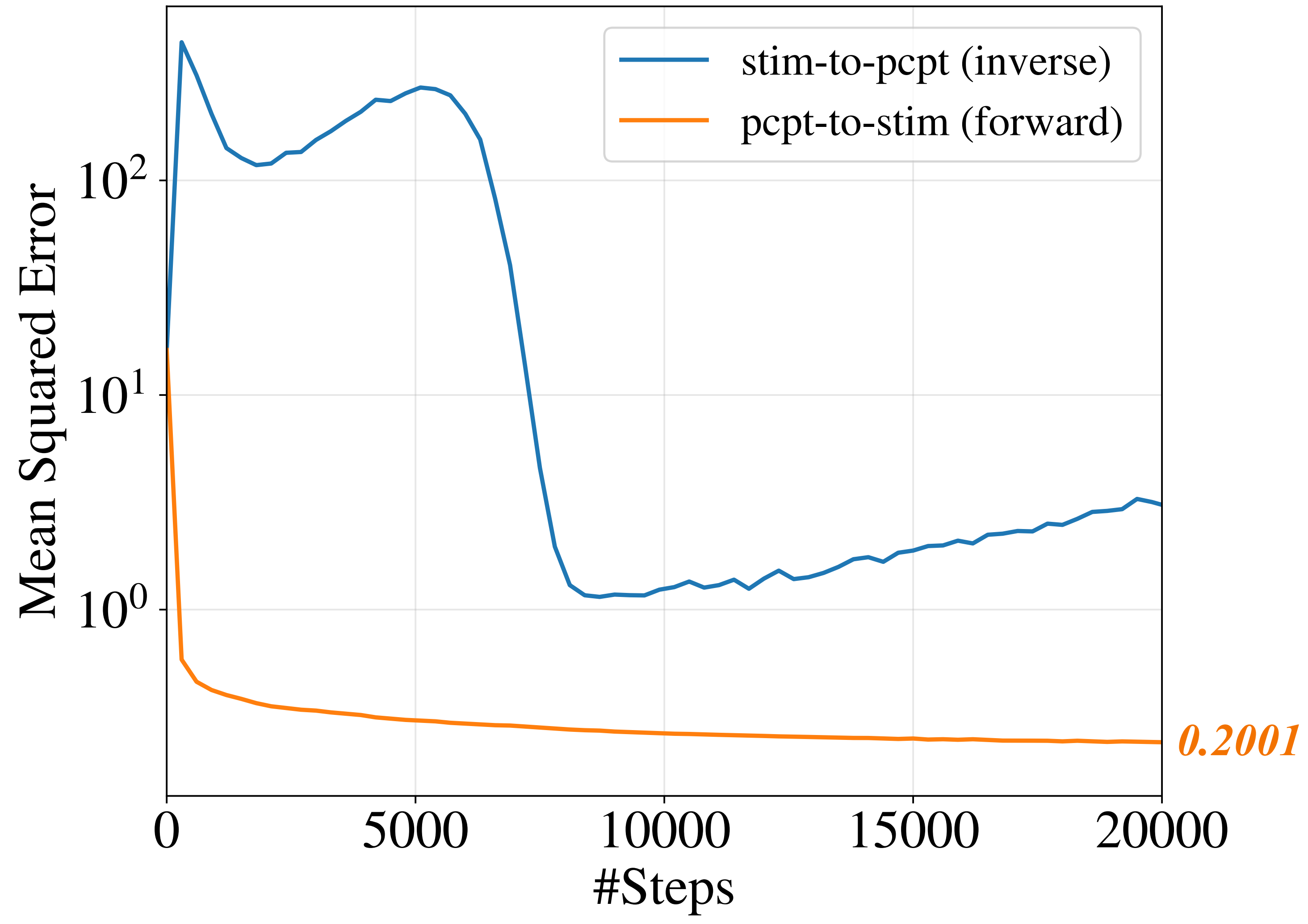

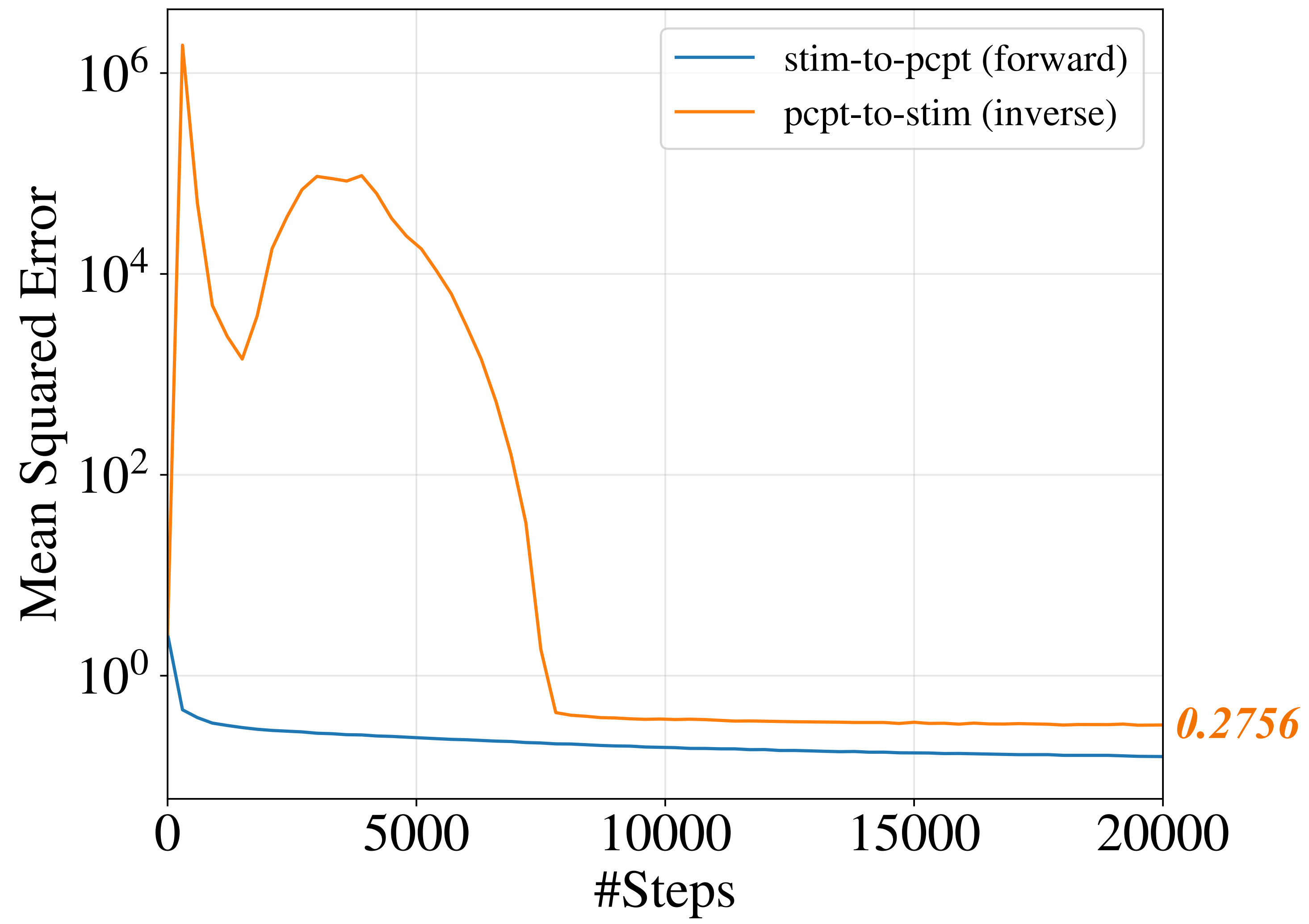

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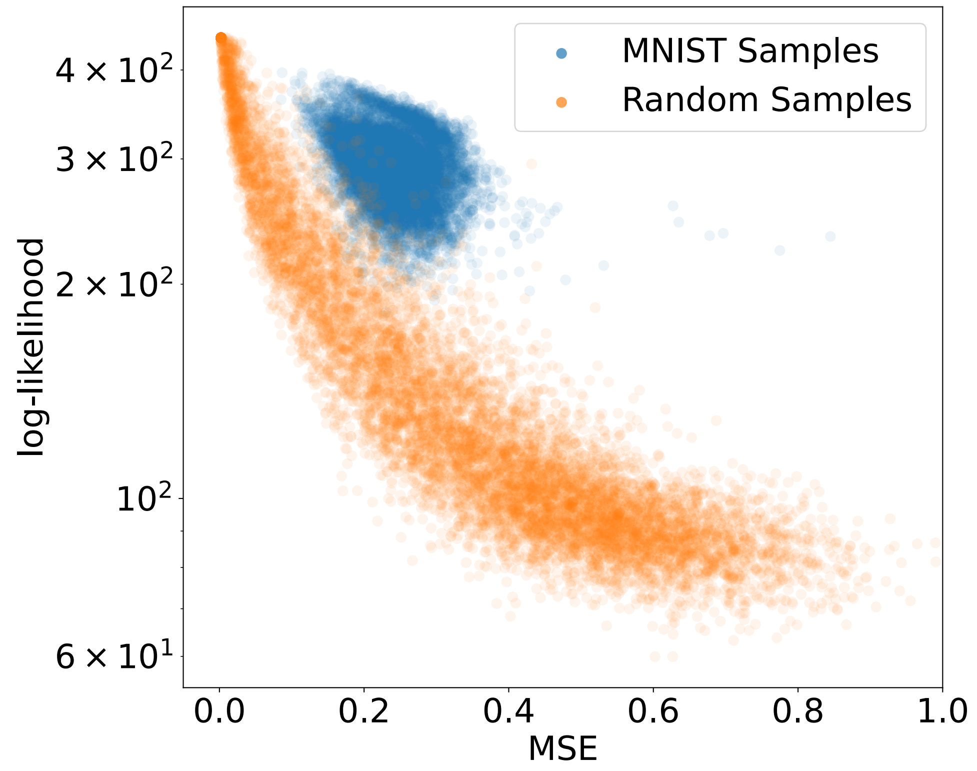
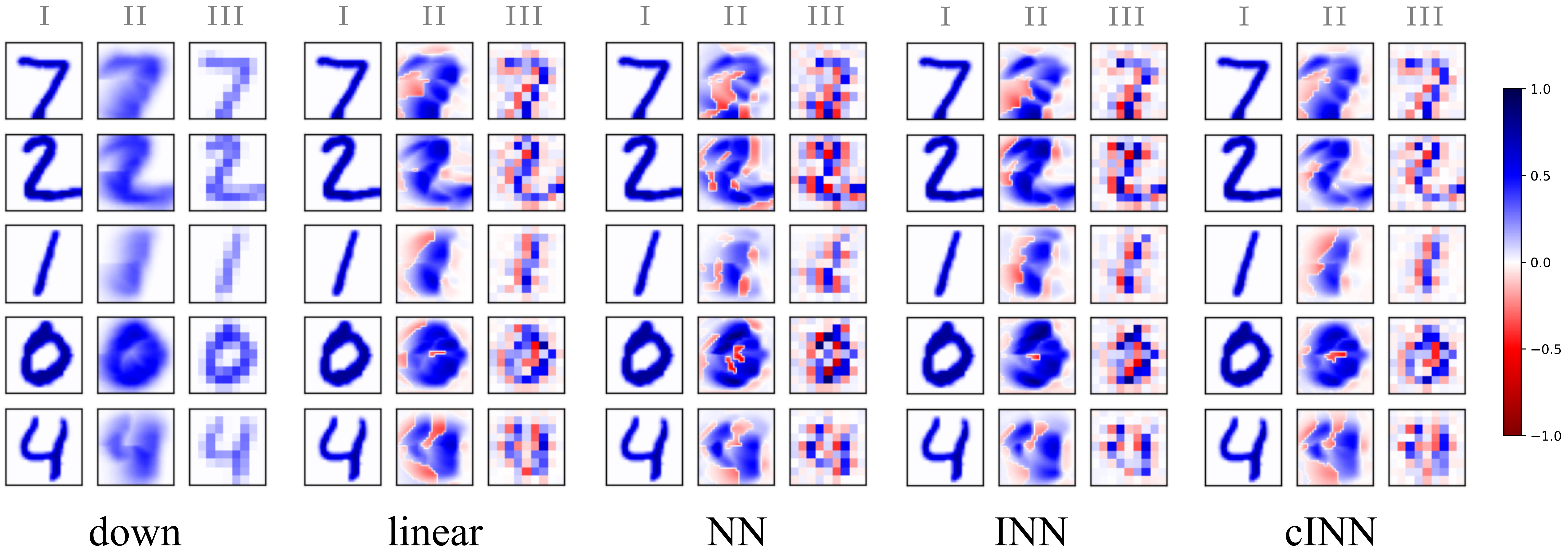
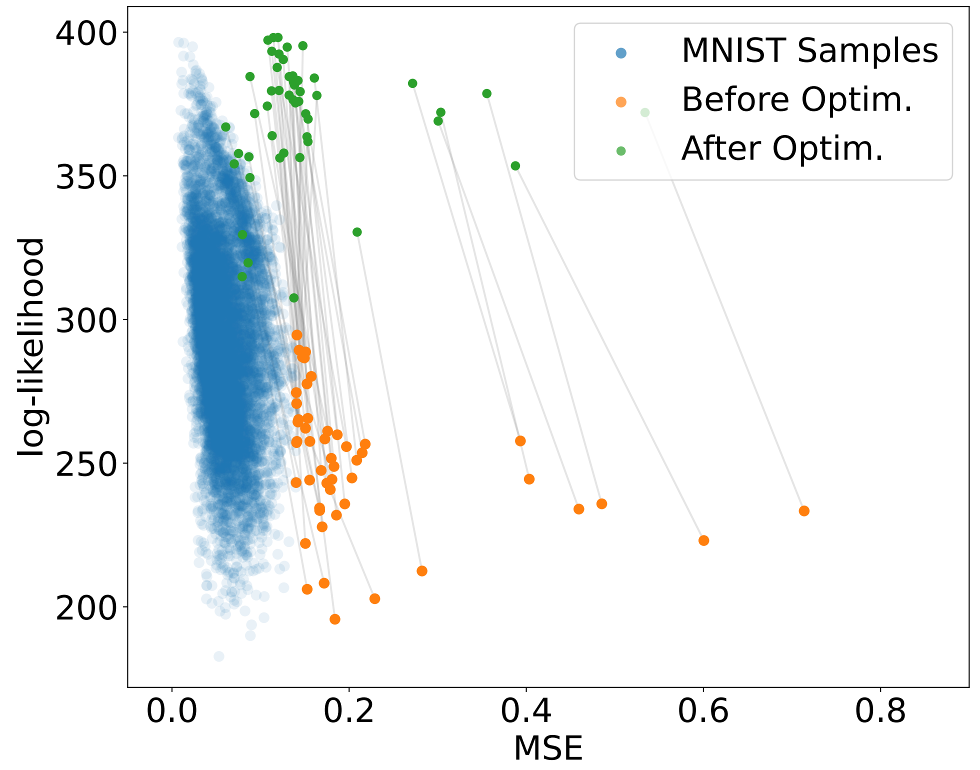
Invertible Neural Networks

Methods	Res.	MAE ↓	SSIM ↑	PSNR ↑	ACC ↑	#Param ↓
down		0.3432	0.3784	25.0973	0.2884	1
linear	28	0.3279	0.4770	25.1453	0.5884	63,504
NN	×	0.3340	0.4901	25.2400	0.6192	4,270,900
INN	28	0.3231	0.4937	25.2729	0.5764	18,063,390
cINN		0.3136	0.4943	25.5161	0.5932	2,449,197







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